LOW POWER VLSI DESIGN USING CLOCK GATING

PROJECT REPORT

Submitted By

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ABSTRACT

- Problem : High dynamic power consumption due to unnecessary clock switching.
- Solution : Implemented clock gating.
- Result : Reduced switching activity and dynamic power.

Objectives

- Understand power consumption in digital circuits.
- Implement clock gating.
- Compare power before and after clock gating.
- Analyze timing impact.

Introduction

<u>1.Dynamic Power :</u>

Dynamic power is the power consumed when transistors switch between logic states (0 and 1). In digital circuits, charging and discharging of load capacitances during signal transitions cause dynamic power consumption. It is the major source of power consumption in synchronous circuits operating at high frequencies.

The dynamic power is given by:

$P_{dynamic} = \alpha C V^2 f$

Where:

α = Switching activity factor

C = Load capacitance

V = Supply voltage

f = Clock frequency

Since the clock signal toggles continuously, it contributes significantly to dynamic power consumption. Reducing unnecessary switching activity helps in lowering dynamic power.

2.Static Power :

Static power is the power consumed even when a circuit is not switching. It is mainly caused by leakage currents flowing through transistors when they are in the OFF state. Although static power was negligible in older technologies, it has become increasingly important in modern deep-submicron VLSI designs.

Sources of static power include:

- Sub-threshold leakage current
- Gate oxide leakage current
- Junction leakage current

Static power is independent of switching activity and exists as long as power is supplied to the circuit.

<u>3.Clock Network Power :</u>

The clock network distributes the clock signal to all sequential elements such as flip-flops and registers. Since the clock toggles every cycle and drives a large number of circuit elements, the clock network consumes a significant portion of the total chip power.

Characteristics of clock network power:

- Continuous switching regardless of circuit activity
- High fan-out to multiple registers
- Large contribution to overall dynamic power

In many digital designs, the clock network alone can consume a considerable percentage of the total dynamic power. Therefore, optimizing clock activity is an effective way to reduce power consumption.

4.Clock Gating Concept :

Clock gating is a low-power design technique used to reduce unnecessary clock switching. It works by disabling the clock signal to inactive portions of a circuit using an enable control signal.

When the enable signal is active, the clock is allowed to pass and the circuit operates normally. When the enable signal is inactive, the clock is blocked, preventing unnecessary switching in registers and sequential logic.

Advantages of clock gating:

- Reduces dynamic power consumption
- Decreases switching activity
- Improves overall power efficiency
- Widely used in modern ASIC and FPGA designs

Clock gating is one of the most effective and commonly used techniques for low-power VLSI design.

5.RTL Clock Gating VS Synthesis Clock Gating

RTL Clock Gating :

In RTL clock gating, the designer manually implements clock gating logic in the Verilog code using logic gates such as AND gates along with an enable signal.

Example:

```
assign gated_clk = clk & enable;
```

Advantages:

- Easy to understand and implement
- Demonstrates clock gating functionality

Disadvantages:

- May introduce glitches
- Can cause timing issues
- Not recommended for industrial ASIC designs

Synthesis Clock Gating :

In synthesis clock gating, the designer writes enable conditions in the RTL code without manually gating the clock. During synthesis, the synthesis tool automatically identifies gating opportunities and inserts Integrated Clock Gating (ICG) cells.

Advantages:

- Glitch-free operation
- Better timing performance
- Industry-standard implementation
- Optimized power reduction

Because of its reliability and efficiency, synthesis clock gating is preferred in professional VLSI design flows.

<u>DESIGN METHODOLOGY</u>

Design Without Clock Gating :

Initially, the design was implemented using a normal clock signal without any power optimization technique. The clock signal was directly connected to the sequential elements, causing the registers to receive clock pulses continuously regardless of whether data processing was required. Functional simulation was performed to verify correct operation of the design. Synthesis and power analysis reports were then generated using Vivado.

Design with Clock Gating:

To reduce unnecessary switching activity, clock gating was introduced into the design. An enable signal was used to control the clock distribution. When the enable signal was active, the clock was allowed to propagate through the circuit. When the enable signal was inactive, the clock was blocked, preventing unnecessary transitions in the sequential elements.

The modified design was simulated to verify functional correctness. Synthesis and power analysis were then performed to evaluate the impact of clock gating on power consumption and overall design performance.

<u>Simulation Process:</u>

The following steps were performed during simulation

1. Creation of Verilog RTL design.
2. Development of a testbench for verification.
3. Functional simulation using Vivado Simulator.
4. Verification of output waveforms.
5. Comparison of behavior before and after clock gating.

<u>Synthesis Process:</u>

The design was synthesized using Vivado synthesis tools to generate the hardware implementation. Synthesis reports were analyzed to study:

- Resource utilization
- Timing performance
- Power consumption
- Design optimization

Power Analysis:

Power analysis was carried out using Vivado power estimation reports. The power consumption of the original design was compared with the clock-gated design. The comparison helped evaluate the effectiveness of clock gating in reducing dynamic power consumption.

Design Flow:

The overall design flow followed in this project is shown below

Specification

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RTL Design in Verilog

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Testbench Development

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Functional Simulation

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RTL Verification

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Synthesis using Vivado

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Power Analysis

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Result Comparison

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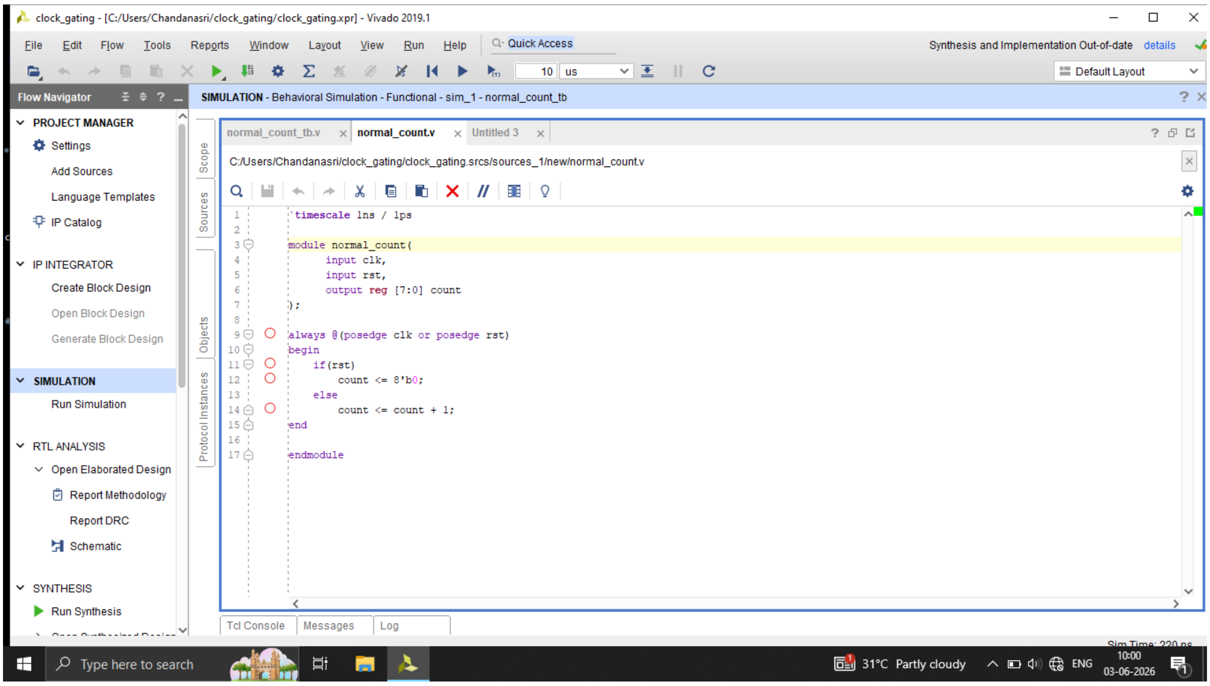
Conclusion

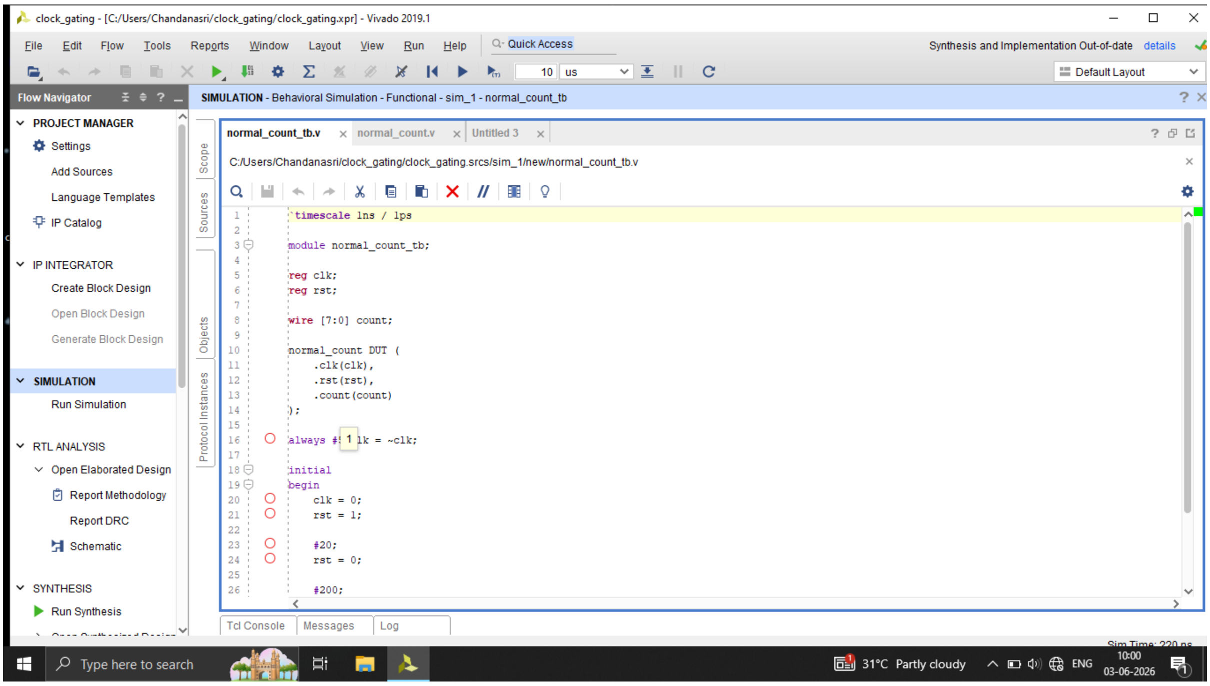
<u>Screenshots of the project:</u>

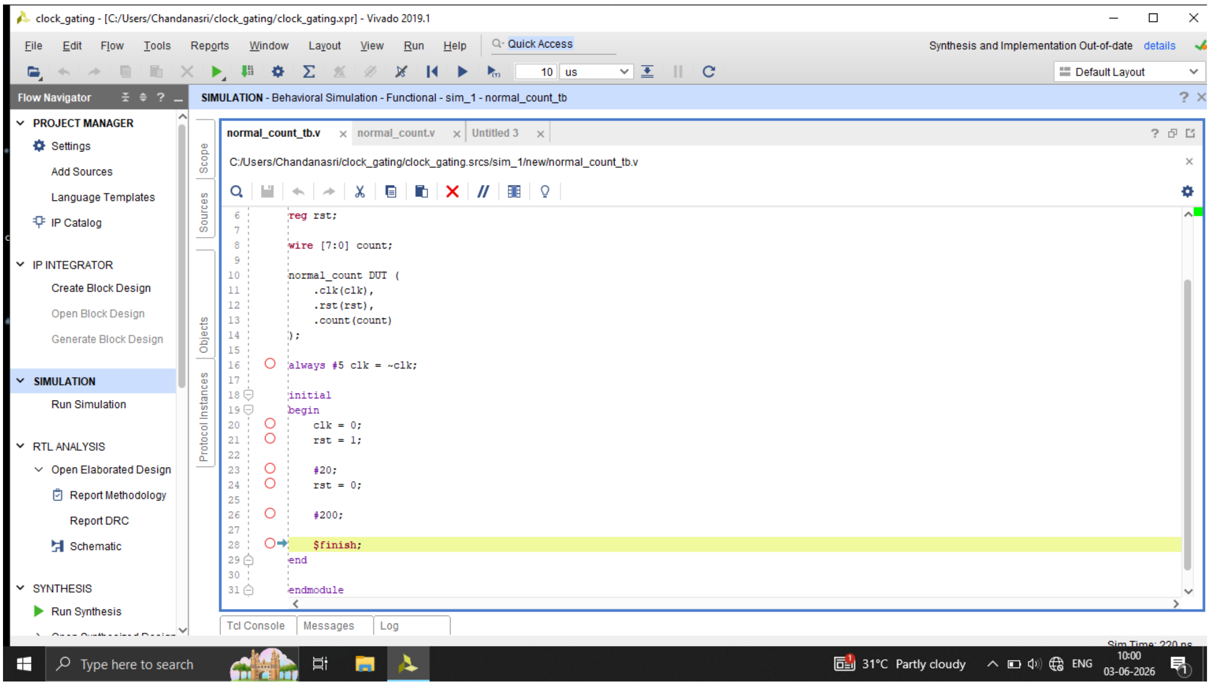
Without Clock Gating

1.RTL Code

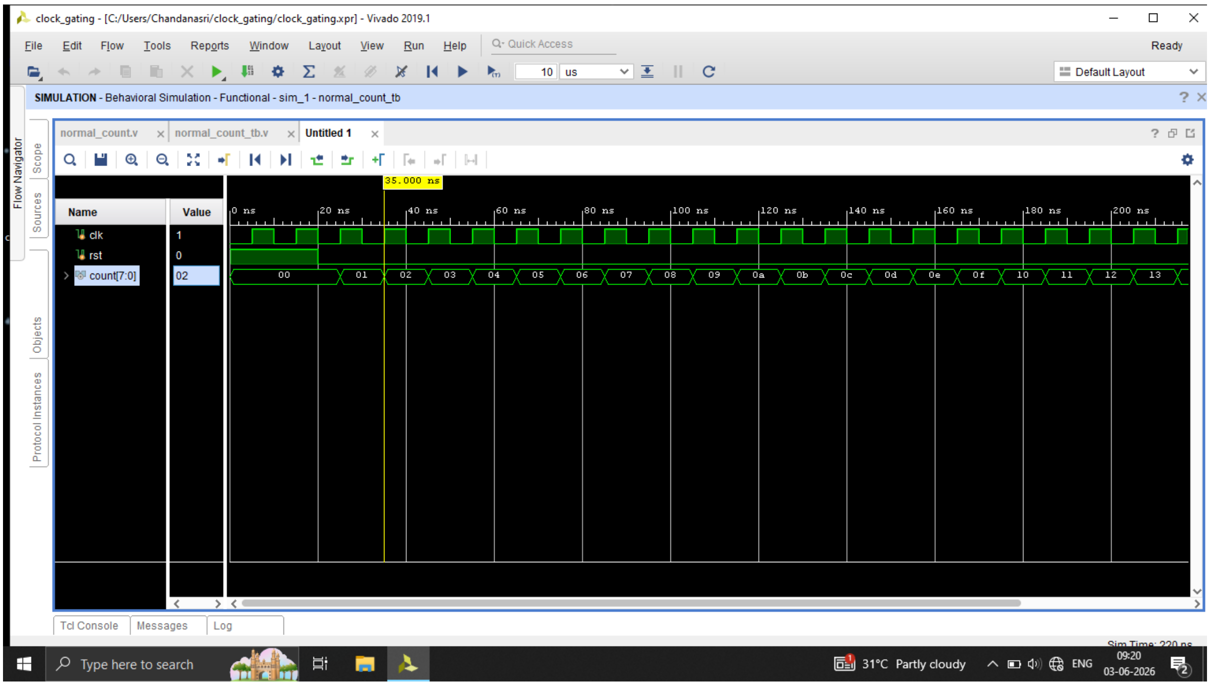
- Design Module
- Testbench Module



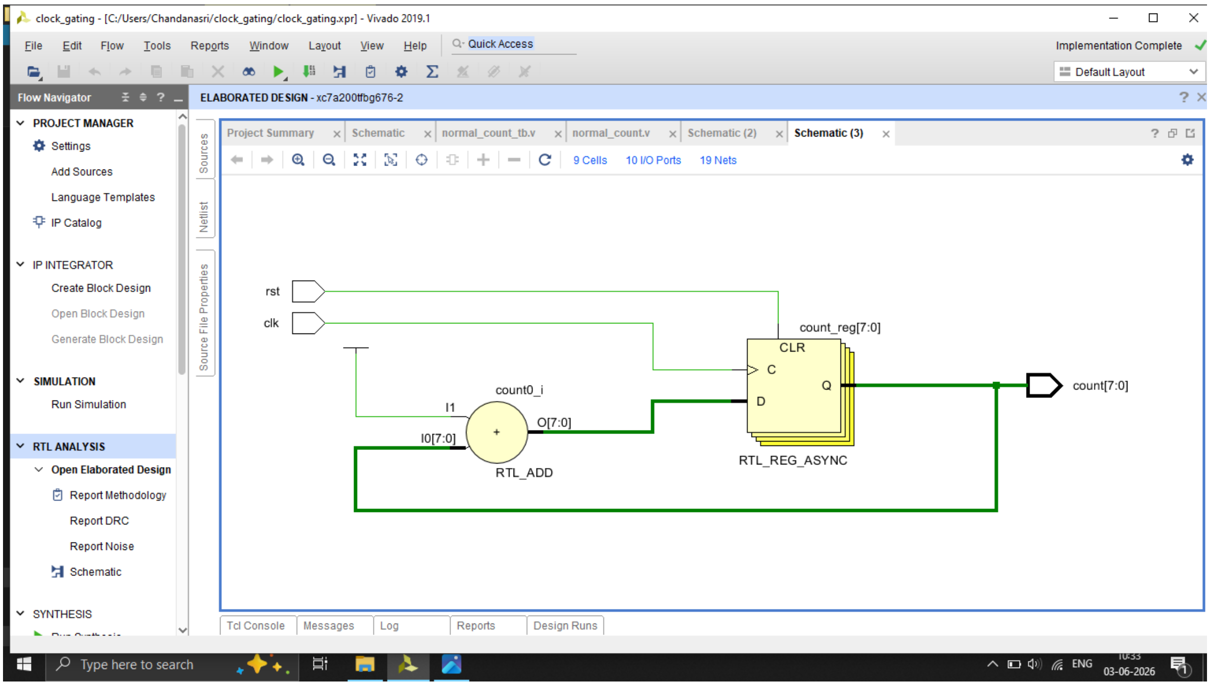




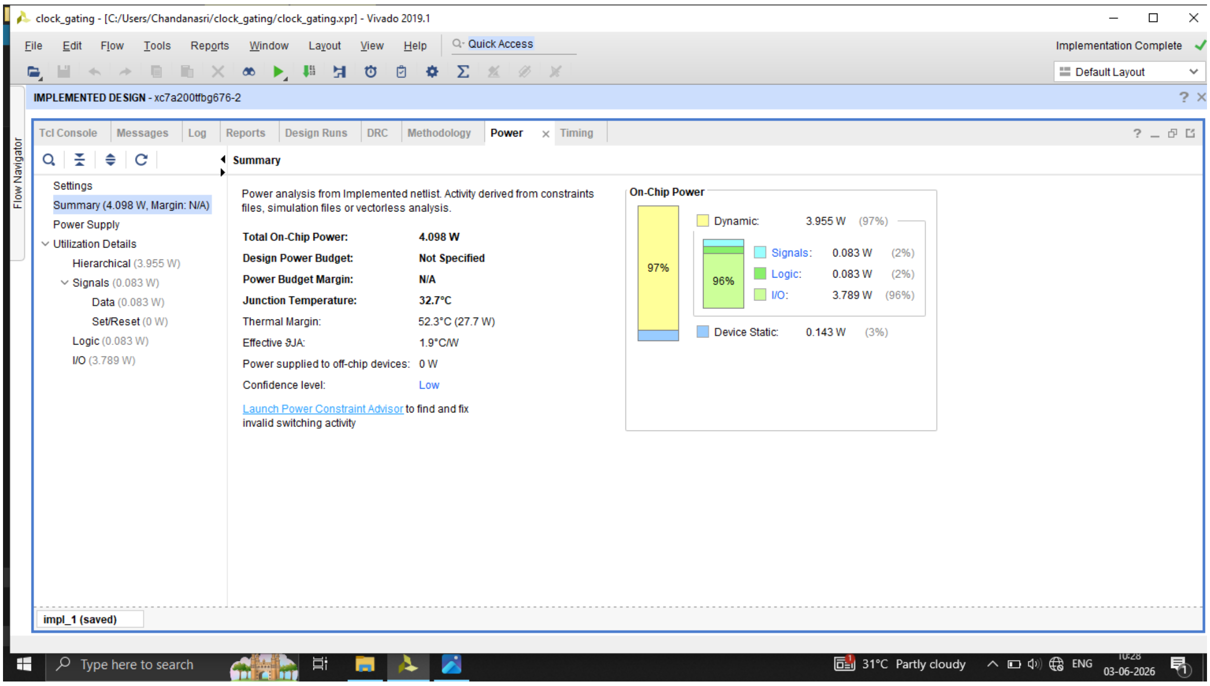
2.Waveform



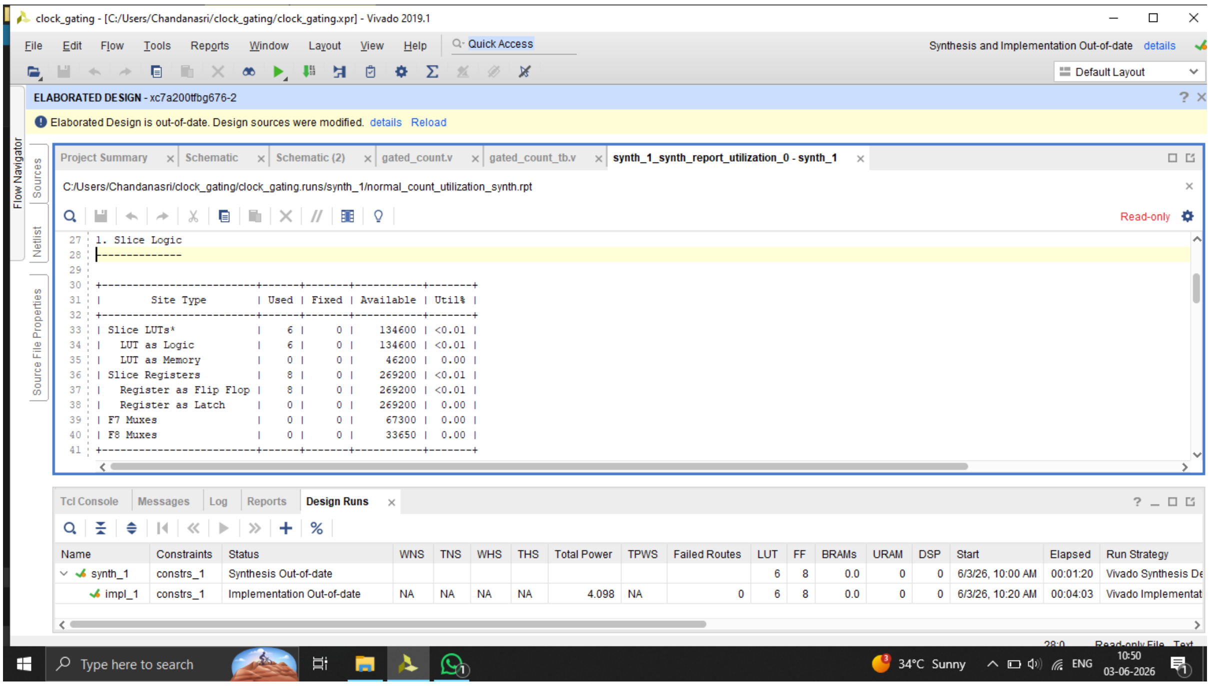
3.Schematic Diagram



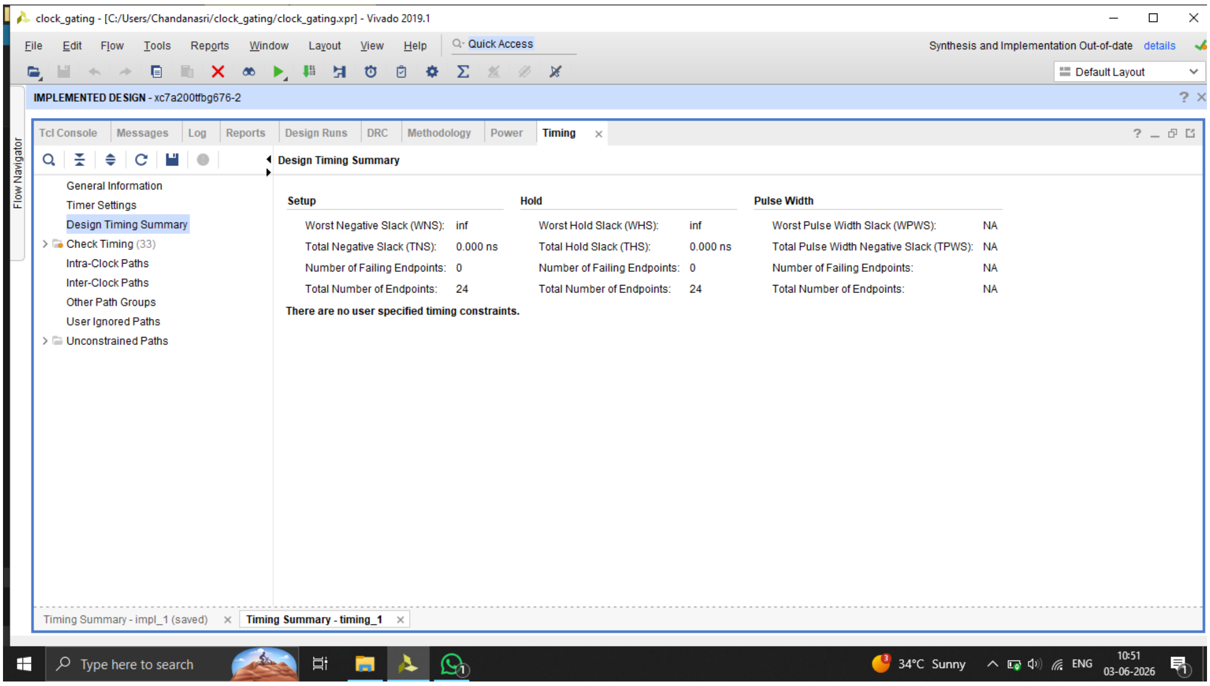
4.Power Report



Utilization Report



Timing Summary

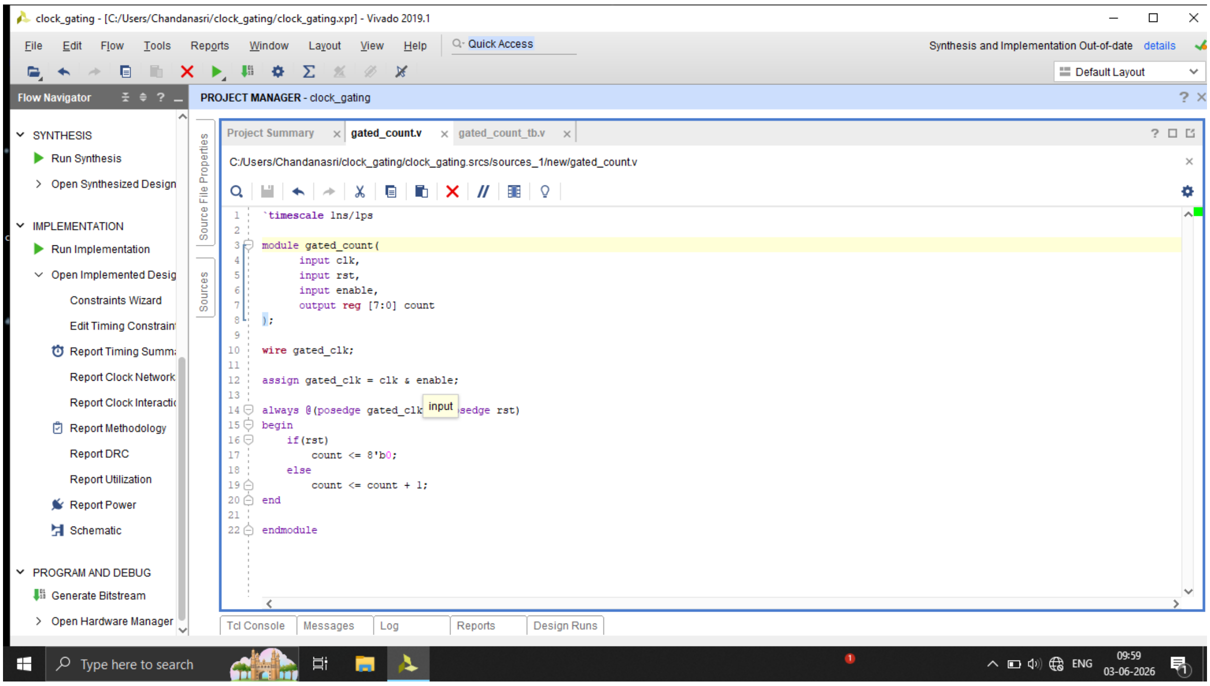


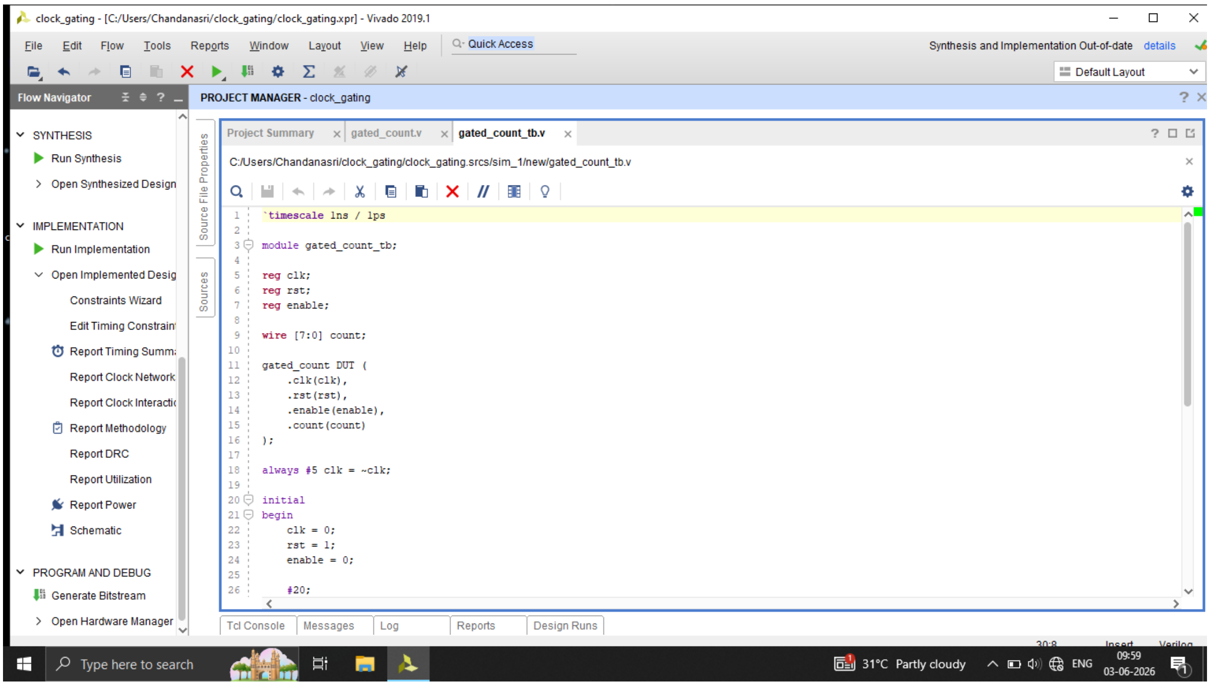
With Clock Gating

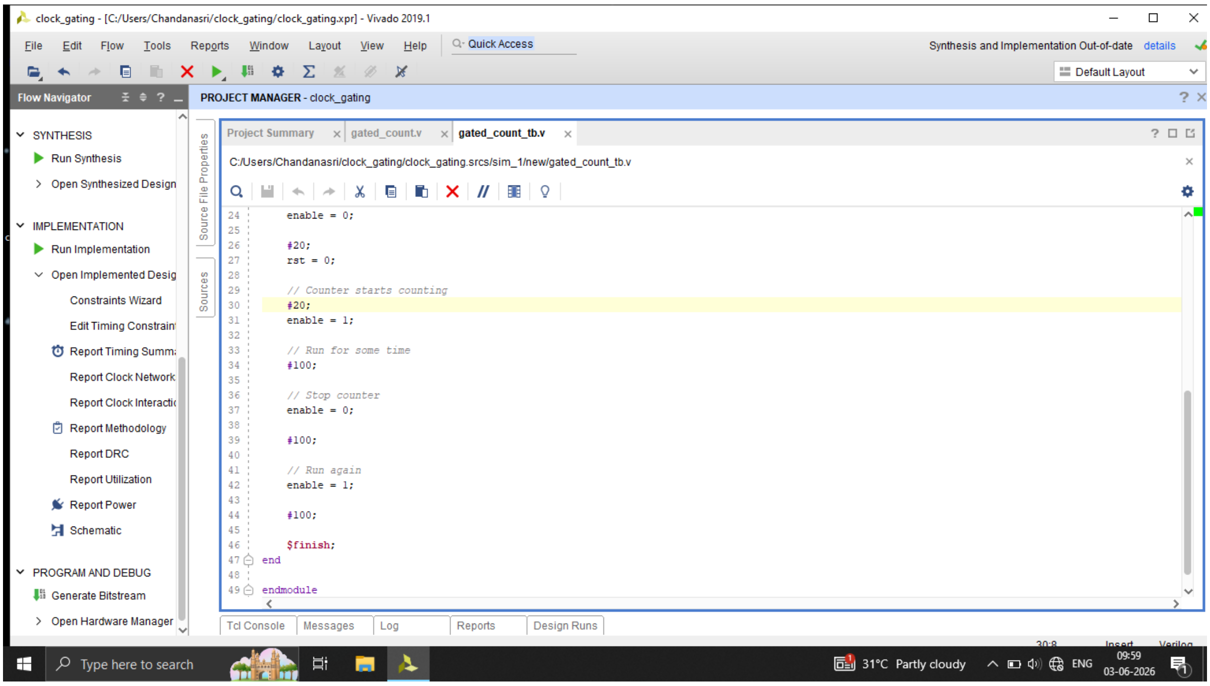
1.RTL code

- Design Module

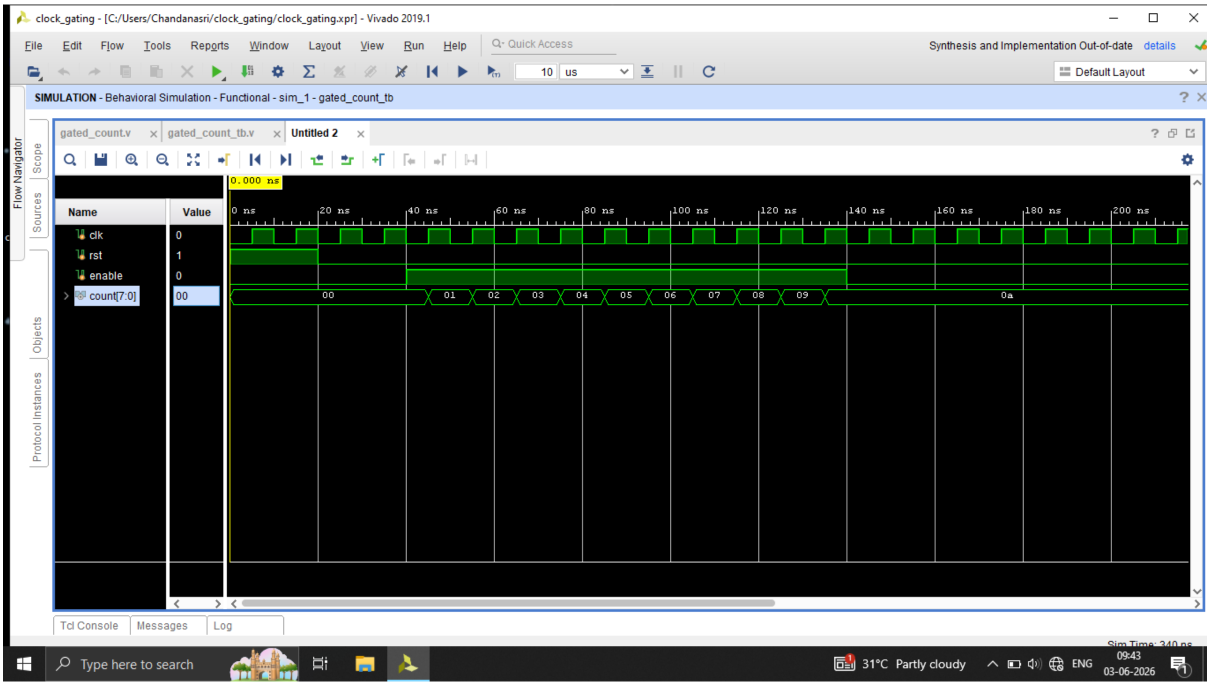
- Testbench Module

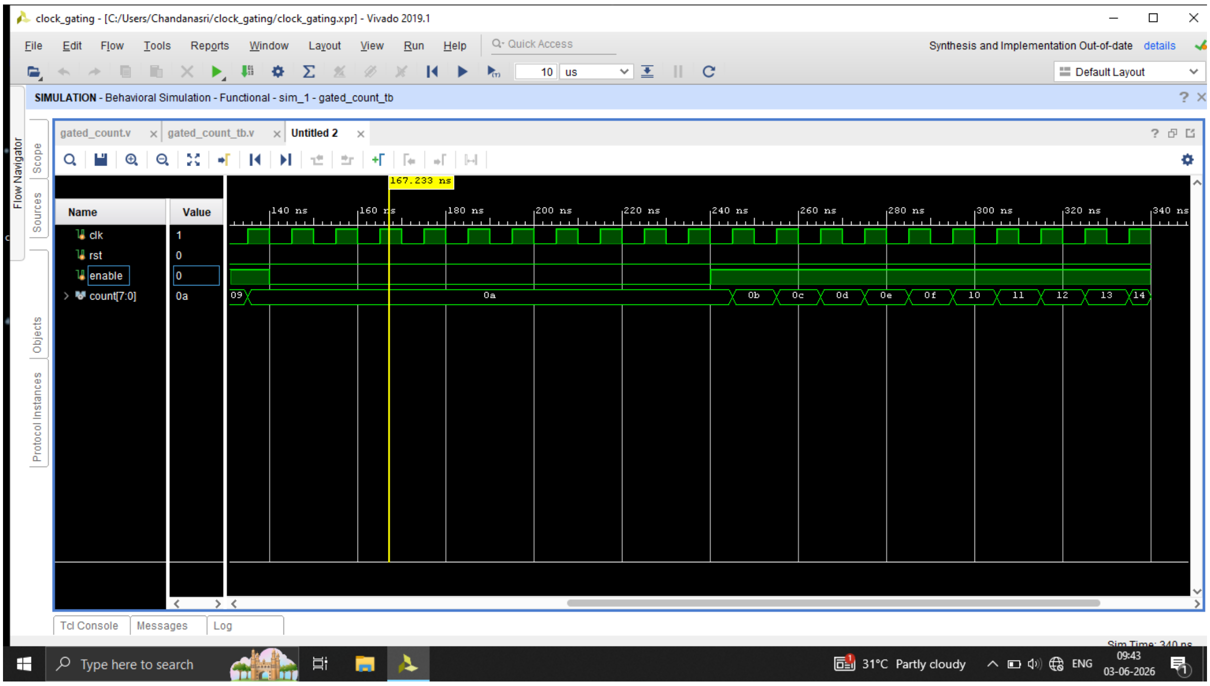




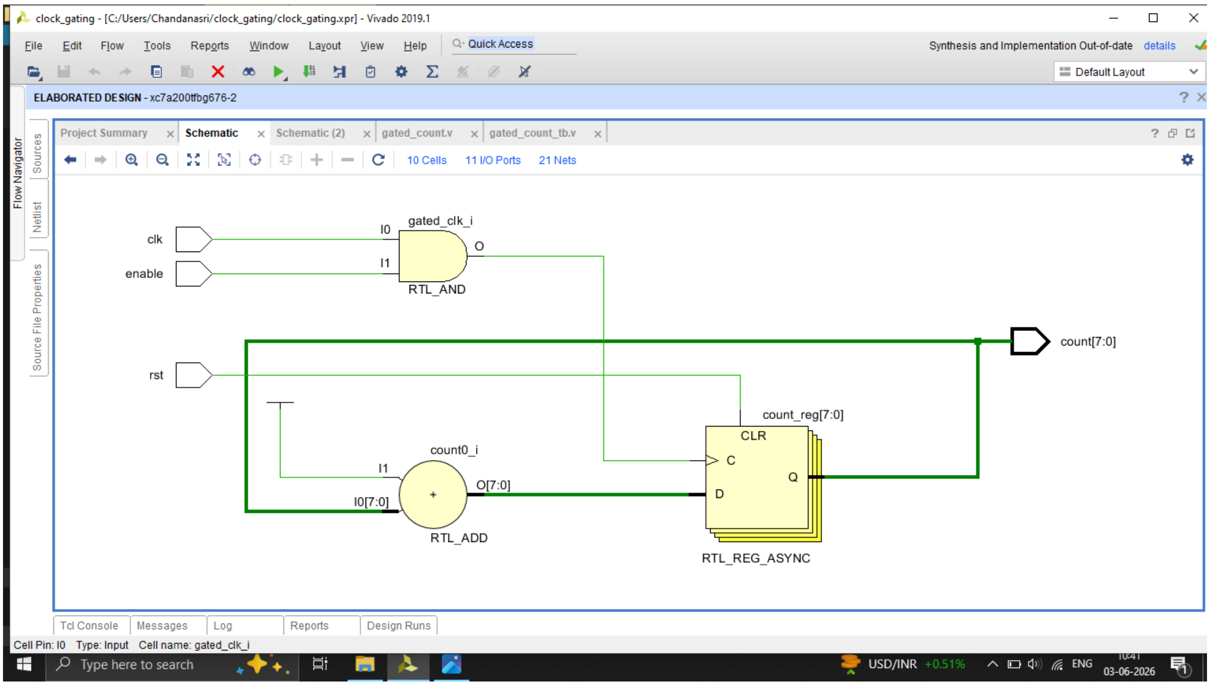

clock_gating - [C:/Users/Chandanasri/clock_gating/clock_gating.xpr] - Vivado 2019.1
PROJECT MANAGER - clock_gating
C:/Users/Chandanasri/clock_gating/clock_gating.srcs/sim_1/new/gated_count_tb.v
enable = 0;
#20;
rst = 0;
// Counter starts counting
#20;
enable = 1;
// Run for some time
#100;
// Stop counter
enable = 0;
#100;
// Run again
enable = 1;
#100;
$finish;
end
endmodule

2.Waveforms

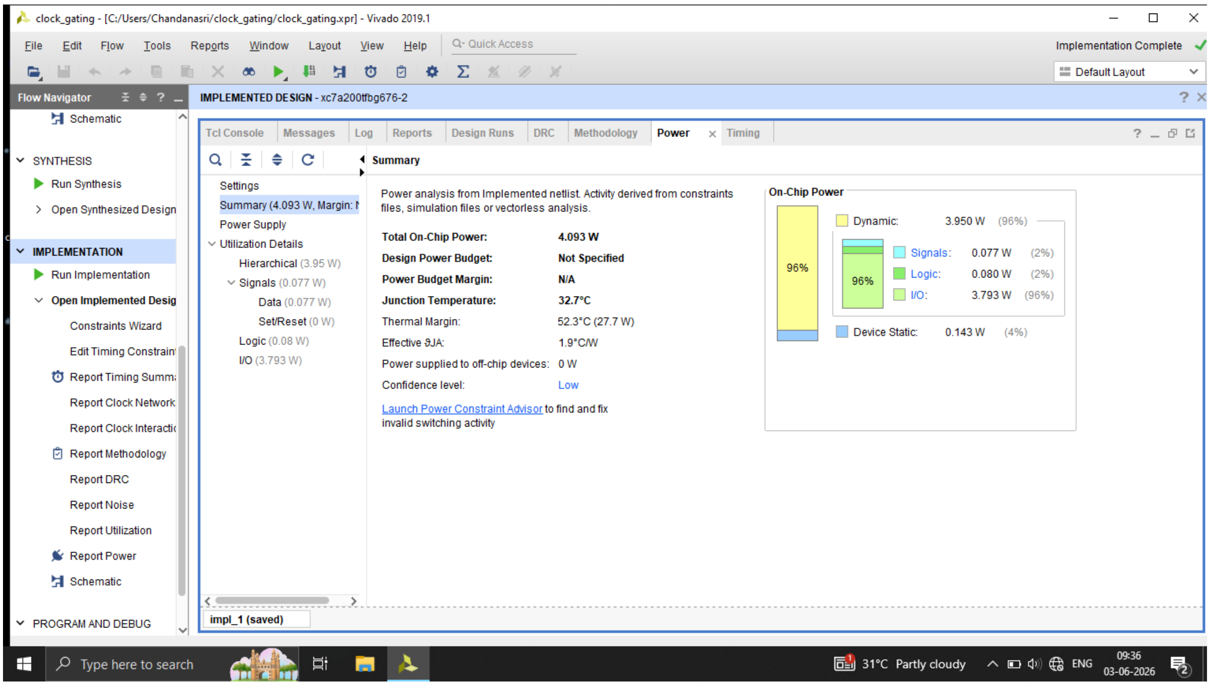

clock_gating - [C:/Users/Chandanasri/clock_gating/clock_gating.xpr] - Vivado 2019.1
SIMULATION - Behavioral Simulation - Functional - sim_1 - gated_count_tb
Name
Value
clk
0
rst
1
enable
0
count[7:0]
00
00
01
02
03
04
05
06
07
08
09
0a



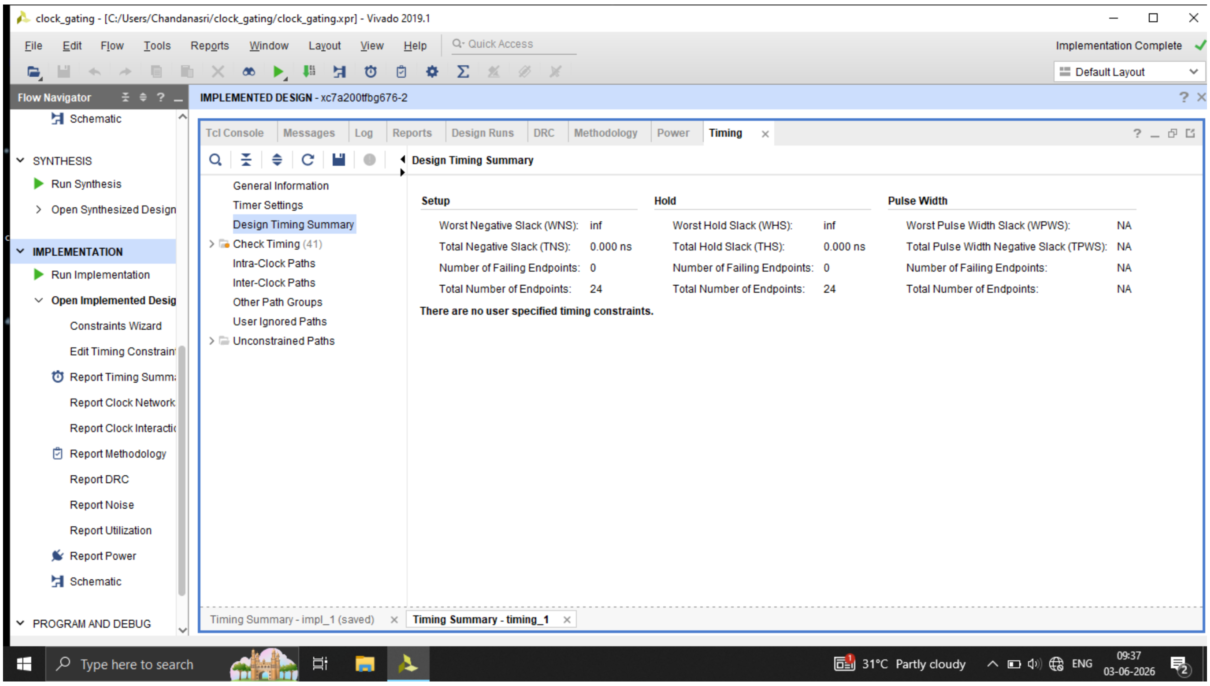
3.Schematic Diagram



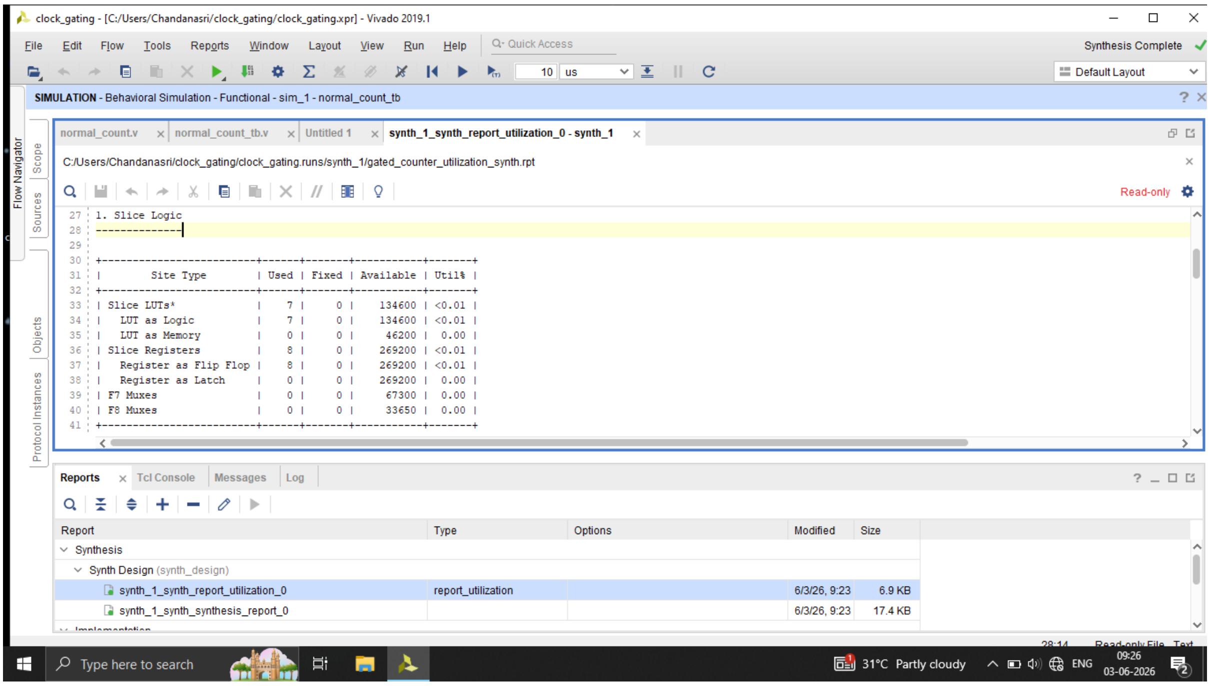
4.Power Report



Timing Summary



Utilization Report



RESULTS AND ANALYSIS

Resource Utilization Analysis:

The synthesis reports show that the normal counter design utilizes 6 LUTs and 8 Flip-Flops. After implementing clock gating, the utilization increased slightly to 7 LUTs while the number of Flip-Flops remained unchanged at 8.

This increase in LUT count is due to the additional logic required for clock gating. However, the area overhead is minimal and acceptable considering the power optimization achieved.

Power Analysis:

Power reports generated using Vivado indicate a reduction in dynamic power consumption after implementing clock gating.

The total on-chip power decreased from 4.098 W to 4.093 W. Dynamic power decreased from 3.955 W to 3.950 W. Signal power reduced from 0.083 W to 0.077 W and logic power reduced from 0.083 W to 0.080 W.

The reduction in dynamic power demonstrates that clock gating successfully reduces unnecessary switching activity within the design.

Timing Analysis:

Timing reports show-

- Worst Negative Slack (WNS): No violations

- Total Negative Slack (TNS): 0 ns
- Number of Failing Endpoints: 0

The design meets timing requirements both before and after clock gating. Therefore, power optimization was achieved without introducing timing violations.

RTL Schematic Analysis:

The RTL schematic of the normal design consists of an 8-bit counter driven directly by the clock signal.

In the clock-gated design, an additional AND gate is introduced between the clock and the counter registers. The enable signal controls whether the clock reaches the sequential elements. When the enable signal is low, clock transitions are blocked, thereby reducing unnecessary switching activity.

Discussion:

The implementation of clock gating successfully reduced switching activity and dynamic power consumption while maintaining correct functionality and timing performance. Although a small increase in logic utilization was observed, the trade-off is acceptable for low-power design applications.

The results validate clock gating as an effective low-power technique commonly used in modern VLSI systems.

CONCLUSION:

This project successfully demonstrated the implementation of a low-power VLSI design using the clock gating technique in Vivado. Two versions of an 8-bit counter were designed and analyzed: a conventional counter and a clock-gated counter.

Simulation results verified the correct functionality of both designs. Synthesis, timing, and power analysis reports were generated and compared. The clock-gated design showed a reduction in switching activity, resulting in lower dynamic power consumption while maintaining correct functionality and timing performance.

The results confirm that clock gating is an effective power optimization technique for reducing unnecessary clock transitions in digital circuits. Although the reduction observed in this project is relatively small due to the simplicity of the design, the same concept provides substantial power savings in large-scale VLSI systems containing thousands of sequential elements.

Overall, this project provided practical experience in RTL design, simulation, synthesis, timing analysis, power analysis, and low-power design methodologies used in modern digital integrated circuits.

FUTURE SCOPE:

The scope of this project can be extended in several ways

1. Implementation of Integrated Clock Gating (ICG) cells in industrial ASIC design flows.
2. Application of clock gating to larger digital systems such as processors, controllers, and communication modules.
3. Investigation of advanced low-power techniques including power gating, multi-voltage design, and dynamic voltage scaling.
4. Comparison of RTL clock gating and synthesis-based clock gating in ASIC design environments.
5. Analysis of power savings in complex designs with a larger number of registers and clock domains.
6. Implementation of the design on FPGA hardware and measurement of practical power consumption.

The study of these advanced techniques can further improve power efficiency and provide deeper insight into modern low-power VLSI design methodologies.

Tools Used:

- Vivado Design Suite 2019.1
- Verilog HDL
- Vivado Simulator
- Vivado Synthesis Tool
- Vivado Power Analyzer
- Vivado Timing Analyzer

REFERENCES:

[1] Xilinx Vivado Design Suite 2019.1 User Guide.

[2] Samir Palnitkar, Verilog HDL: A Guide to Digital Design and Synthesis, Pearson Education.

[3] Neil H. E. Weste and David Harris, CMOS VLSI Design: A Circuits and Systems Perspective, Pearson Education.

[4] Course notes and online documentation related to Clock Gating and Low-Power VLSI Design.